

limb? That information is provided by spatial patterns of *Hox* genes, which specify different developmental fates in particular body regions (see Figure 21.20).

BMP-4, FGF, Sonic hedgehog, and *Hox* proteins are examples of a much larger set of molecules that govern cell fates in animals. Having mapped out many of the basic functions of these molecules in embryonic development, researchers are now addressing their role in organogenesis, focusing in particular on the development of the brain.

Cilia and Cell Fate

Researchers have found that the cellular organelles known as cilia are essential for specifying cell fate in human embryos. Like other mammals, humans have stationary and motile cilia (see Figure 6.24). Stationary primary cilia, or *monocilia*, jut from the surface of nearly all cells, one per cell. In contrast, motile cilia are restricted to cells that propel fluid over their surface, such as the epithelial cells of airways, and sperm (as flagella that propel sperm movement). Both stationary and motile cilia have crucial roles in development.

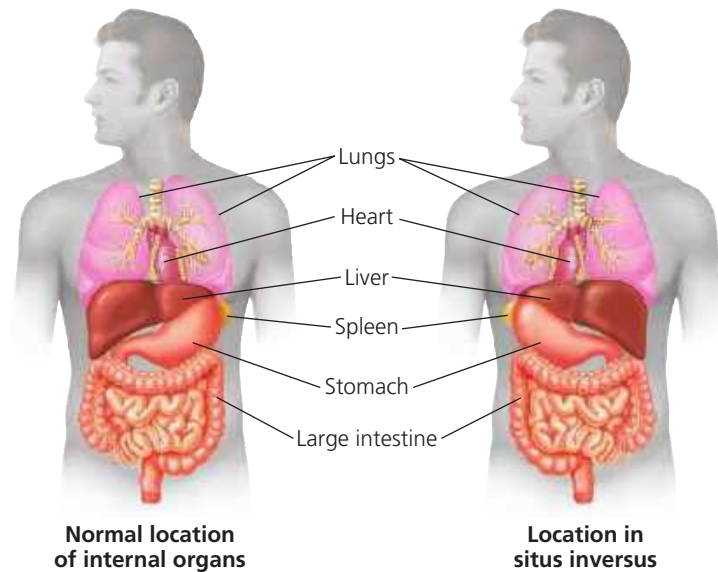
Genetic studies provided vital clues to the developmental role of monocilia. In 2003, researchers discovered that certain mutations disrupting development of the mouse nervous system affect genes that function in the assembly of monocilia. Other geneticists found that mutations responsible for a severe kidney disease in mice alter a gene important for the transport of materials up and down monocilia. In addition, mutations in humans that block the function of monocilia were linked to cystic kidney disease.

How do monocilia function in development? Evidence indicates that monocilia act as antennae on the cell surface, receiving signals from multiple signaling proteins, including Sonic hedgehog. Mechanisms that regulate which types of receptor proteins are present tune the cilium to particular signals. When the monocilia are defective, signaling is disrupted.

Insight into the role of motile cilia in development grew from studies of Kartagener's syndrome, a set of medical conditions that often appear together. These conditions include male infertility due to immotile sperm and infections of the nasal sinuses and bronchi in both males and females. A distinctive feature of Kartagener's syndrome is *situs inversus*, a reversal of the normal left-right asymmetry of the organs in the chest and abdomen (Figure 47.27). The heart, for example, is on the right side rather than the left. (By itself, *situs inversus* causes no significant medical problems.)

Scientists studying Kartagener's syndrome came to realize that all of the associated conditions result from a defect that makes cilia immotile. Without motility, sperm tails cannot beat and airway cells cannot sweep mucus and microbes out

▼ **Figure 47.27** *Situs inversus*, a reversal of normal left-right asymmetry in the chest and abdomen.



of the airway. But what causes *situs inversus* in these individuals? The current model proposes that ciliary motion in a particular part of the embryo is essential for normal development. Evidence indicates that movement of the cilia generates a leftward fluid flow, breaking the symmetry between left and right sides. Without that flow, asymmetry along the left-right axis arises randomly, and half of the affected embryos develop *situs inversus*.

If we consider development as a whole, we see a sequence of events marked by cycles of signaling and differentiation. Initial cell asymmetries allow different types of cells to influence each other, resulting in the expression of specific sets of genes. The products of these genes then direct cells to differentiate into specific types. Through pattern formation and morphogenesis, differentiated cells ultimately produce a complex arrangement of tissues and organs, each functioning in its appropriate location and in coordination with other cells, tissues, and organs throughout the organism.

CONCEPT CHECK 47.3

1. How do axis formation and pattern formation differ?
2. **MAKE CONNECTIONS** How does a morphogen gradient differ from cytoplasmic determinants and inductive interactions with regard to the set of cells it affects (see Concept 18.4)?
3. **WHAT IF?** If the ventral cells of an early frog gastrula were experimentally induced to express large amounts of a protein that inhibits BMP-4, could a second embryo develop? Explain.
4. **WHAT IF?** If you removed the ZPA from a limb bud and then placed a bead soaked in Sonic hedgehog in the middle of the bud, what would be the most likely result?

For suggested answers, see Appendix A.